Superconductivity and superfluidity

Cool certain metals to a few degrees above absolute zero, and their electrical resistance vanishes completely, and magnetic fields are pushed out. Cool helium, and it becomes a liquid without friction that creeps up glass walls and flows through holes nothing else can pass. Both phenomena are quantum mechanics on a large scale: billions of particles behaving as a single wave. This page tells the story from Kamerlingh Onnes to today's quantum computers – with two interactive figures.

The quest for cold

Helium was the last gas that had not been liquefied. On 10 July 1908 Heike Kamerlingh Onnes succeeded in Leiden: helium boils at 4.2 K, and for fifteen years Leiden was the only place in the world where such low temperatures could be reached. Onnes used the helium to measure the resistance of metals. It was known that resistance falls as the temperature falls – but did it go to zero, or rise again because the electrons froze in place?

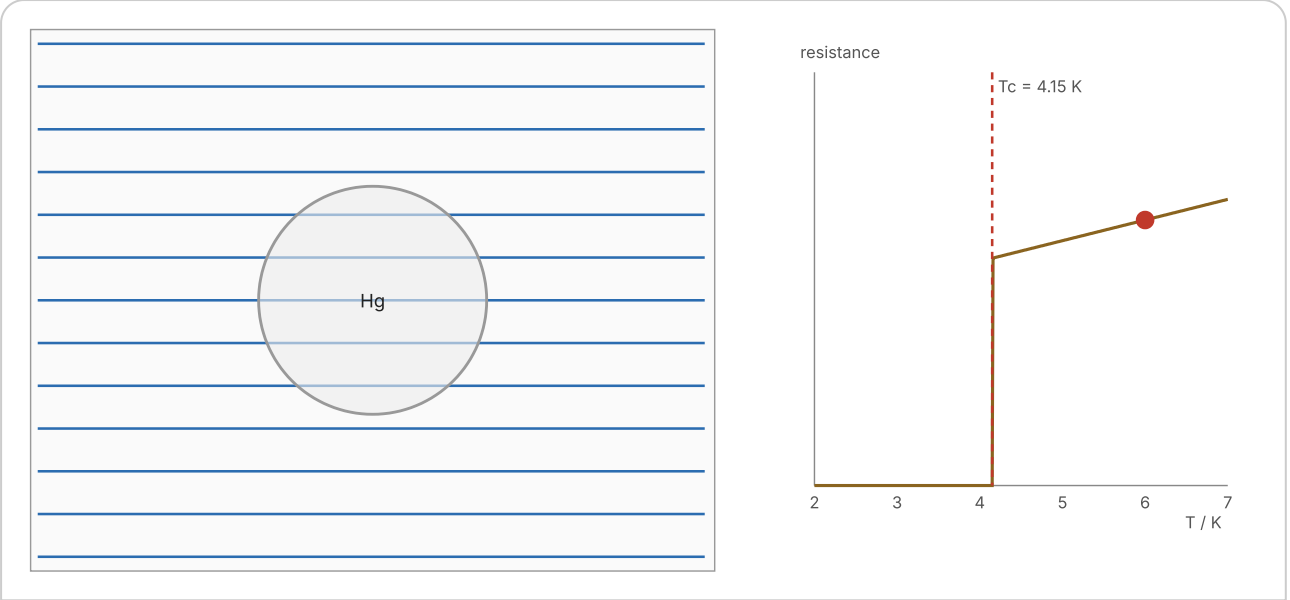
In 1911 Onnes and his assistant Gilles Holst measured mercury. At 4.2 K the resistance suddenly dropped below what could be measured. It was not a gradual development but a jump – a new state of matter. Onnes called it *supraconductivity*. Later experiments have shown that a current in a superconducting ring can flow for years without measurable decay. Onnes received the Nobel Prize in 1913.

The Meissner effect: more than zero resistance

A superconductor is not just a perfect conductor. In 1933 Walther Meissner and Robert Ochsenfeld discovered that a superconductor actively expels a magnetic field when it is cooled below the critical temperature T_c – even if the field was there before it was cooled. A perfect conductor, by contrast, would simply freeze in whatever field was inside it. The Meissner effect shows that superconductivity is a genuine thermodynamic phase, and it is why a magnet can levitate above a superconductor.

In 1935 the brothers Fritz and Heinz London explained the effect with an equation that makes the field decay exponentially into the superconductor over a penetration depth λ of typically 50–500 nm:

$$\nabla^2 \vec{B} = \frac{\vec{B}}{\lambda^2}, \qquad \lambda = \sqrt{\frac{m}{\mu_0 n_s e^2}}$$



Cooper pairs and the BCS theory

For 46 years superconductivity was one of the great puzzles of physics. An important clue came in 1950: different isotopes of mercury have different T_c, and T_c is inversely proportional to the square root of the atomic mass. So the vibrations of the crystal lattice had to be involved.

In 1956 Leon Cooper showed that two electrons can form a bound pair if they attract each other, however weakly. The attraction comes from the lattice: a passing electron pulls the positive ions slightly towards itself, and this small concentration of positive charge attracts a second electron. In 1957 John Bardeen, Cooper and Robert Schrieffer worked out the full theory (BCS). All the Cooper pairs in the metal take part in one common quantum state. Breaking a pair costs an energy, the gap 2Δ, and at low temperatures there is not enough energy to scatter the pairs. The current flows without resistance. BCS theory predicts

$$\Delta(0) \approx 1.764\, k_B T_c, \qquad T_c \propto M^{-1/2}$$

Bardeen, Cooper and Schrieffer received the Nobel Prize in 1972. Bardeen had already received it for the transistor – he is the only person to have won the physics prize twice.

Vortices and flux quanta

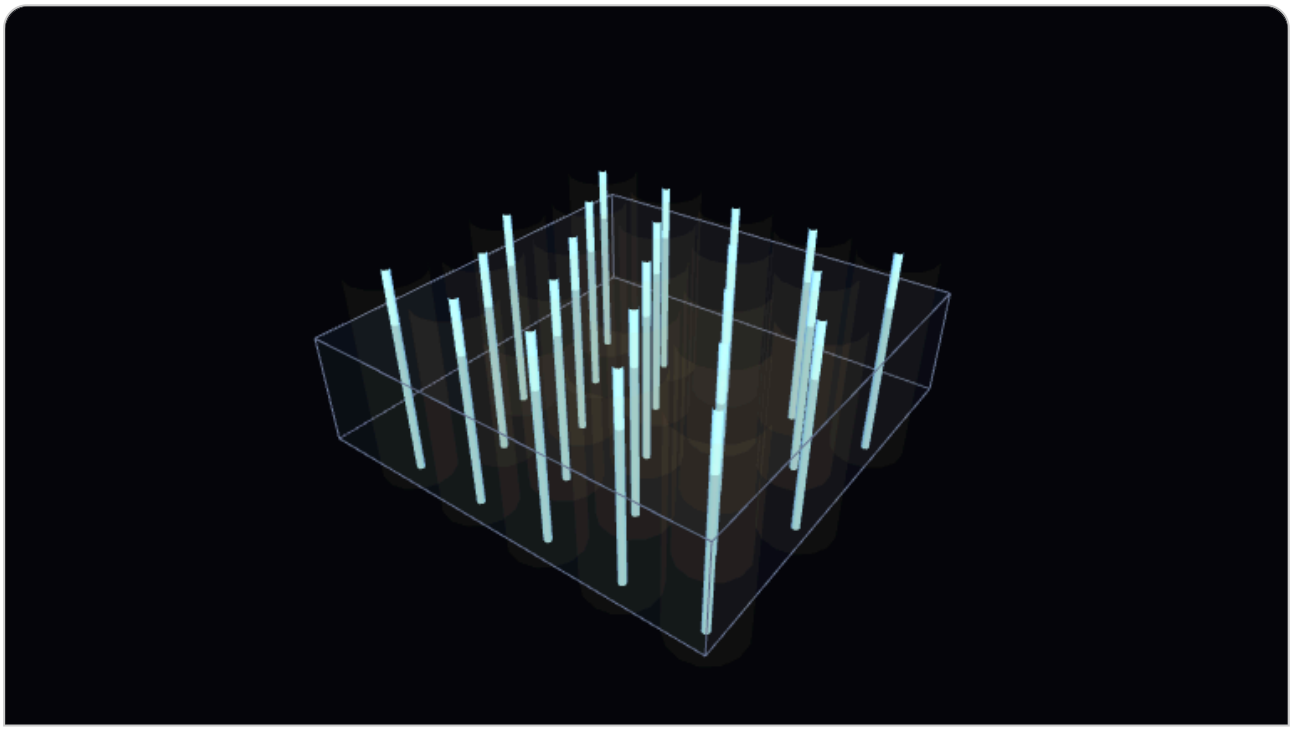
As early as 1950 Vitaly Ginzburg and Lev Landau had formulated a theory of superconductors with a complex "order parameter" ψ, a kind of macroscopic wave function. In 1957 Alexei Abrikosov showed that the theory predicts two types of superconductor. In type I the field is expelled completely until superconductivity breaks down. In type II the field penetrates in thin vortex lines – tubes in which superconductivity is destroyed, surrounded by circulating currents –

while the rest remains superconducting. Almost all technical superconductors are type II. Ginzburg and Abrikosov shared the Nobel Prize in 2003 with Anthony Leggett.

Because the wave function must be single-valued round a vortex, the magnetic flux in each vortex is quantised. Measurements in 1961 showed that the flux quantum contains the charge $2e$ – direct evidence for the electron pairs:

$$\Phi_0 = \frac{h}{2e} \approx 2.07 \cdot 10^{-15}\ \mathrm{Wb}$$

Abrikosov's vortex lattice (3D)



Josephson and the SQUID

In 1962 Brian Josephson, a 22-year-old PhD student in Cambridge, predicted that Cooper pairs can tunnel through a thin insulating barrier between two superconductors with no voltage drop, and that a DC voltage V across the barrier gives an alternating current of frequency $f = 2eV/h$. Both were quickly confirmed. Today the Josephson effect is used to define the volt: 1 μV corresponds to 483.6 MHz. He received the Nobel Prize in 1973.

A SQUID (superconducting quantum interference device) is a superconducting ring with two Josephson junctions. It can measure magnetic fields of less than a billionth of the Earth's field and is used to record the magnetic signals of the brain. Superconducting circuits with Josephson junctions are also the basis of the qubits that IBM and Google use in their quantum computers – a development resting on Leggett's theory of macroscopic quantum mechanics.

High-temperature superconductors

For 75 years the highest critical temperatures stayed below 25 K, and many believed that BCS theory set a limit around 30 K. In 1986 Georg Bednorz and Alex Müller at IBM in Zürich found superconductivity at 35 K in a ceramic copper oxide. The following year Paul Chu and Maw-Kuen Wu found that $YBa_2Cu_3O_7$ becomes superconducting at 93 K – above the boiling point of liquid nitrogen, 77 K, which is cheap and easy to handle. Bednorz and Müller received the Nobel Prize in 1987, faster than anyone before them.

The mechanism behind superconductivity in the cuprates is still not fully understood after almost forty years of research. Since then new families have been found: magnesium diboride (39 K, 2001), iron-based superconductors (2008) and hydrides under enormous pressure, e.g. H_3S at 203 K (2015) and LaH_{10} at about 250 K (2019) – but only at pressures of over a million atmospheres. Claims of superconductivity at room temperature and normal pressure have so far not been confirmed; in 2023 two widely reported papers in *Nature* were retracted, and the material LK-99 turned out not to be superconducting.

Applications

MRI scanners

The magnets in hospital MRI scanners are coils of niobium–titanium, cooled with liquid helium to 4 K. They give fields of 1.5–3 tesla – and are the largest application of superconductors today.

Accelerators

The LHC at CERN has 1232 superconducting dipole magnets of 8.3 tesla, cooled to 1.9 K with superfluid helium. See Particle physics and group theory.

Fusion

New fusion reactors use tapes of high-temperature superconductor (REBCO) for magnets of up to 20 tesla, so that the tokamak can be made much smaller.

Maglev trains

The Japanese SCMaglev levitates on superconducting magnets and in 2015 set the world speed record for trains at 603 km/h.

Superfluid helium

Helium-4 never solidifies at normal pressure – the zero-point motion is too large. But at 2.17 K something happens. The heat capacity, which Willem Keesom measured in 1927, has a sharp peak shaped like the Greek letter λ, and the transition is called the λ point. Below it, helium no longer boils with bubbles but stands completely still.

In January 1938 two short papers appeared side by side in *Nature*: Pyotr Kapitsa in Moscow and John Allen and Don Misener in Cambridge had independently found that helium below the λ point flows through ultrafine slits with no measurable viscosity. Kapitsa called it *superfluidity*. The same year Allen and Harry Jones discovered the fountain effect: heat helium in a tube with a porous plug, and it squirts out in a fountain. Superfluid helium also creeps in a thin film up the walls of its container and drips off the bottom on the outside.

In 1938 Fritz London proposed that superfluidity is a Bose–Einstein condensation: a macroscopic fraction of the atoms gathers in the lowest quantum state. His calculation gave a transition temperature of 3.1 K – close to 2.17 K, even though helium atoms interact strongly. László Tisza and Lev Landau developed a two-fluid model: below the λ point helium behaves as a mixture of a normal liquid and a superfluid component with no entropy and no viscosity. The model predicted a new type of wave in which the two components oscillate in antiphase – “second sound”, a temperature wave – which Vasily Peshkov measured in 1944.

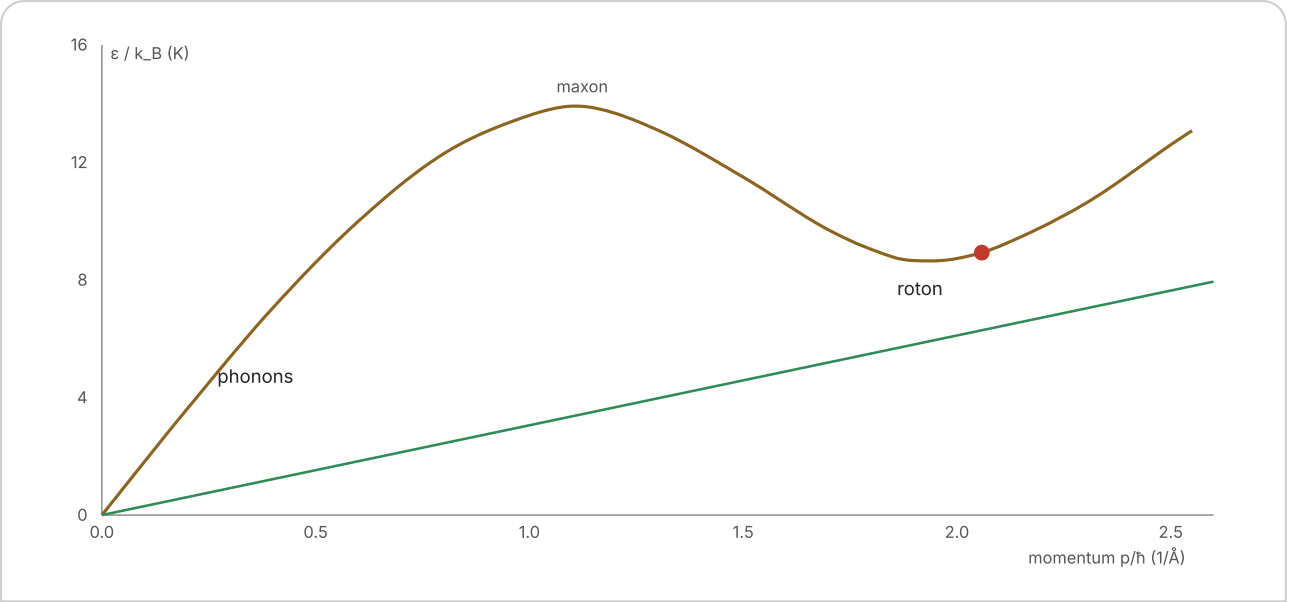
Landau’s critical velocity

Why does superfluid helium feel no friction? Landau gave an elegant argument in 1941. Friction means that the liquid loses energy by creating small excitations – sound quanta (phonons) or other motions – as it flows past a wall. An excitation with momentum p and energy $\varepsilon(p)$ can only be created if the flow velocity exceeds

$$v_c = \min_p \frac{\varepsilon(p)}{p}$$

For an ordinary gas $\varepsilon = p^2/2m$, and $v_c = 0$: there is always friction. But in helium the excitations have a special spectrum, with sound at small momenta and a minimum – the *roton* – at

larger ones. Neutron-scattering experiments have since measured the curve precisely. The smallest slope from the origin to the curve gives Landau's critical velocity, about 60 m/s.



Lars Onsager (1949) and Richard Feynman (1955) realised that rotation in a superfluid can only take place through quantised vortices, in which the circulation is a whole multiple of h/m. William Vinen measured the quantum of circulation in 1961. Landau received the Nobel Prize in 1962, Kapitsa only in 1978.

Helium-3 and ultracold atomic gases

Helium-3 atoms are fermions and cannot Bose-condense directly. But in 1972 Douglas Osheroff, David Lee and Robert Richardson found that helium-3 too becomes superfluid, at about 2 mK – because the atoms form pairs, just like the electrons in a superconductor. There are several superfluid phases, and at Sussex Anthony Leggett explained their strange magnetic properties as a spontaneously broken symmetry between spin and motion. The discoverers received the Nobel Prize in 1996, Leggett in 2003.

In 1995 Eric Cornell and Carl Wieman in Boulder and Wolfgang Ketterle at MIT succeeded in making Bose–Einstein condensates in dilute gases of rubidium and sodium atoms at about 100 nK. Ten years later superfluid gases of fermionic atoms were produced and lattices of quantised vortices observed. Ultracold gases are today a laboratory in which the interactions can be controlled and superconductors imitated.

Superfluidity also occurs in nature: the neutrons in the interior of neutron stars are thought to be superfluid, and sudden jumps in the rotation of pulsars ("glitches") are explained by the release of vortices in the superfluid.

The common principle

Superconductivity and superfluidity are two sides of the same coin. In both, a macroscopic number of particles – Cooper pairs or helium atoms – is in one common quantum state that can be described by a single wave function $\psi = \sqrt{n}\, e^{i\varphi}$. The flow velocity is determined by the gradient of the phase, $\vec{v}_s = \frac{\hbar}{m}\nabla\varphi$, and because the phase must be single-valued, circulation and magnetic flux are quantised. In the superconductor the particles are charged, so the current is tied to the magnetic field – which gives the Meissner effect.

Both phenomena break a symmetry spontaneously: the choice of phase. The same principle – spontaneous symmetry breaking with an order parameter – is the basis of the Higgs mechanism in particle physics; as early as 1962 Philip Anderson pointed out that the photon in a superconductor in effect acquires a mass. See Forces from symmetry.

The Bose–Einstein temperature for an ideal gas of density n and particle mass m. For helium-4 with density 0.145 g/cm³ it gives 3.1 K

$$T_{\mathrm{BE}} = \frac{2\pi\hbar^2}{m k_B}\left(\frac{n}{\zeta(3/2)}\right)^{2/3}$$

The Josephson frequency

$$f = \frac{2e}{h}V \approx 483.6\ \mathrm{MHz}/\mu\mathrm{V}$$

Further reading

- Dirk van Delft: *Freezing Physics. Heike Kamerlingh Onnes and the Quest for Cold* (2007)
- Michael Tinkham: *Introduction to Superconductivity* (2nd ed., Dover 2004)
- A.J. Leggett: *Quantum Liquids* (Oxford 2006)
- Sébastien Balibar: “The discovery of superfluidity”, *Journal of Low Temperature Physics* 146 (2007)
- Stephen Blundell: *Superconductivity. A Very Short Introduction* (Oxford 2009)

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